



Celestine

△ Crystallized blue celestine

Celestine often forms beautiful, transparent, light to medium-blue crystals – if it were harder and more durable, it might be one of the world’s favourite gemstones. It takes its name from the Latin *coelestis*, meaning “heavenly”, an allusion to its “heavenly” sky-blue crystals. Because celestine is soft and easily broken, it is faceted only for collectors and museums by skilled lapidaries. Single crystals are sometimes sold as pendants, but they are too fragile for general wear. Facet-grade material is found in Namibia and Madagascar.

Specification

Chemical name	Strontium sulphate	Formula	SrSO ₄
Colours	Colourless, red, green, blue	Structure	
Orthorhombic	Hardness 3–3.5	SG 4.0	RI
1.62–1.64	Lustre	Vitreous, pearly on cleavage	
Streak White	Locations	USA, Namibia, Madagascar	

Sulphur groundmass



Crystals on sulphur groundmass | Rough | This group of very light blue celestine crystals is growing at all angles off a sulphur groundmass.

Iron oxide



Fine crystals | Rough | These small but perfectly crystallized celestine crystals have formed on a sheet of the iron oxide mineral limonite.

Prism face



“Heavenly” crystals | Rough | These stunning dark blue crystals from Madagascar live up fully to the name celestine, derived from the Latin for “heavenly”.

Colour fades to clear



Double-terminated crystal | Rough | This unusual bi-coloured celestine crystal is double-terminated, meaning it has termination faces on both ends.

Numerous small facets



Mixed-cut gem | Cut | Celestine is another of the extremely hard-to-cut collector’s gems; this mixed cut shows the very high skill of the cutter.

Prism face

Celestine crystals



Calcite banding

Banded barian celestine | Rough | This variety, barian celestine, is rich in barium (see box, left). Here crystals have grown alongside sphalerite and calcite.